INFO8006 Introduction to Artificial Intelligence

Exercise session 6: Machine learning

Recap

Problem statement

Let $\mathbf{d} \sim p(\mathbf{x}, y)$ be a dataset of N example input-output pairs

$$\mathbf{d} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)\},$$

where $\mathbf{x}_i \in \mathbb{R}^d$ are d-dimensional vectors representing the input values and $y_i \in \mathcal{Y}$ are the corresponding output values. From this data, we want to identify a probabilistic model $p_\theta(y \mid \mathbf{x})$ that best explains the data. The problem is a *regression* problem if $y \in \mathbb{R}$ and a *classification* problem if $y \in \{0, 1, \ldots, C-1\}$.

Linear regression

Linear regression models $p(y \mid \mathbf{x}) = \mathcal{N}(y \mid \mathbf{w}^T\mathbf{x} + b, \sigma^2)$. Its maximum likelihood estimate minimizes the sum of squared errors $\sum_{i=1}^{N}(y_i - (\mathbf{w}^T\mathbf{x}_i + b))^2$. Appending a constant feature 1 to $\mathbf{x}$ to absorb b into $\mathbf{w}$, it is $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$, where $\mathbf{X}$ stacks the inputs $\mathbf{x}_i^T$ and $\mathbf{y}$ the outputs y_i.

Multilayer perceptron

A hidden layer with q hidden units computes

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b}),$$

where $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ is the input vector, $\mathbf{W} \in \mathbb{R}^{q \times d_{\text{in}}}$ is the *weight matrix*, $\mathbf{b} \in \mathbb{R}^q$ is the *bias vector* and σ is an activation function. A network with a single hidden layer is a *shallow network*. Layers composed in series form a deep network with K hidden layers,

$$\begin{aligned}
\mathbf{h}_0 &= \mathbf{x} \\
\mathbf{h}_1 &= \sigma(\mathbf{W}_1\mathbf{h}_0 + \mathbf{b}_1) \\
&\vdots \\
\mathbf{h}_K &= \sigma(\mathbf{W}_K\mathbf{h}_{K-1} + \mathbf{b}_K) \\
\mathbf{y} &= \sigma(\mathbf{W}_y\mathbf{h}_K + \mathbf{b}_y),
\end{aligned}$$

where $\mathbf{W}_k \in \mathbb{R}^{q_k \times q_{k-1}}$ is the weight matrix and $\mathbf{b}_k \in \mathbb{R}^{q_k}$ is the bias vector of the k-th layer. The resulting function $f(\mathbf{x}; \theta)$, where θ denotes the set of all model parameters, is a *multilayer perceptron*.

Activation functions

Activation functions $\sigma : \mathbb{R} \to \mathbb{R}$ introduce non-linearities in the network. Common activation functions include

- the sigmoid $\sigma(x) = \frac{1}{1+\exp(-x)}$, which maps inputs to the range $(0, 1)$;
- the hyperbolic tangent $\sigma(x) = \tanh(x) = \frac{\exp(x)-\exp(-x)}{\exp(x)+\exp(-x)}$, which maps inputs to the range $(-1, 1)$;
- the rectified linear unit $\sigma(x) = \mathrm{ReLU}(x) = \max(0, x)$.

Output layers

The output layer $\mathbf{y} = \sigma(\mathbf{W}_y \mathbf{h}_K + \mathbf{b}_y)$ depends on the task:

- for regression, its width is the dimensionality d_{out} of the output and its activation function is the identity $\sigma(\cdot) = \cdot$;
- for binary classification, its width is 1 and its activation function is the sigmoid, which models the probability $P(Y = 1 \mid \mathbf{x})$;
- for classification into C classes, its width is C and its activation function is the *softmax*

$$\mathrm{Softmax}(\mathbf{z})_i = \frac{\exp(z_i)}{\sum_{i'=0}^{C-1} \exp(z_{i'})} \quad \text{for } i = 0, \ldots, C-1,$$

  so that $\mathbf{y}$ is a vector of C probabilities $P(Y = i \mid \mathbf{x})$ summing to 1.

Loss functions

The parameters θ of a deep network $f(\mathbf{x}; \theta)$ are learned by minimizing a loss function $\mathcal{L}(\theta)$ over a dataset $\mathbf{d} = \{(\mathbf{x}_j, \mathbf{y}_j)\}$ of N input-output pairs. The loss function is derived from the likelihood:

- for regression, assuming a Gaussian likelihood, the loss is the *mean squared error*

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{(\mathbf{x}_j, \mathbf{y}_j) \in \mathbf{d}} (\mathbf{y}_j - f(\mathbf{x}_j; \theta))^2;$$

- for classification, assuming a categorical likelihood, the loss is the *cross-entropy*

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{(\mathbf{x}_j, \mathbf{y}_j) \in \mathbf{d}} \sum_{i=1}^{C} y_{ij} \log f_i(\mathbf{x}_j; \theta).$$

Gradient descent

To minimize a loss function $\mathcal{L}(\theta)$, *gradient descent* updates the model parameters iteratively using the update rule

$$\theta_{t+1} = \theta_t - \alpha \nabla_\theta \mathcal{L}(\theta_t),$$

where θ_0 are the initial parameters of the model and α is the *learning rate*.

Backpropagation

The gradient $\nabla_\theta \mathcal{L}(\theta)$ is computed for a training pair $(\mathbf{x}, \mathbf{y})$ in two passes through the network:

1. the *forward pass* computes all intermediate values of the network, from the input $\mathbf{x}$ up to the prediction $f(\mathbf{x}; \theta)$, and the loss;
2. the *backward pass* applies the chain rule from the loss back to the input, reusing the intermediate values of the forward pass, to obtain the partial derivative of the loss with respect to every parameter.

Universal approximation theorem

The *universal approximation theorem* states that any continuous function can be approximated arbitrarily well by a shallow network with sufficiently many hidden units.

Convolution

For the 1d input $\mathbf{x} \in \mathbb{R}^W$ and the convolutional kernel $\omega \in \mathbb{R}^w$, the discrete *convolution* $\mathbf{x} \circledast \omega$ is a vector of size $W - w + 1$ such that

$$(\mathbf{x} \circledast \omega)[i] = \sum_{m=0}^{w-1} \mathbf{x}_{m+i} \omega_m.$$

Padding the input with p zeros on each side results in an output of size $W - w + 1 + 2p$. For the 3d input tensor $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ with C channels and the 3d convolutional kernel $\omega \in \mathbb{R}^{C \times h \times w}$, the 2d convolution $\mathbf{x} \circledast \omega$ is a 2d tensor of size $(H - h + 1) \times (W - w + 1)$ such that

$$(\mathbf{x} \circledast \omega)[j, i] = \sum_{c=0}^{C-1} \sum_{n=0}^{h-1} \sum_{m=0}^{w-1} \mathbf{x}_{c,n+j,m+i} \omega_{c,n,m}.$$

Convolutional neural networks

A *convolutional layer* is defined by a set of K kernels ω_k of size $C \times h \times w$. It applies the 2d convolution to the input tensor $\mathbf{x}$ of size $C \times H \times W$ to produce a set of K feature maps $\mathbf{o}_k$. With a pooling area of size $h \times w$, *max-pooling* maps a tensor $\mathbf{x} \in \mathbb{R}^{C \times (rh) \times (sw)}$ to a tensor $\mathbf{o} \in \mathbb{R}^{C \times r \times s}$ by keeping the maximum of each area. A *convolutional neural network* (CNN) uses convolutional layers, pooling layers and fully connected layers.

Exercises

In session: exercises 1, 4, 7.

Exercise 1 Escape game (Exam, January 2022)

A new virtual escape game came out, and you decide to play it. You arrive in a 5×5 grid world where each cell (x, y) is a room with doors leading to the adjacent rooms. The game's goal is to reach the exit room as fast as possible, but its position is unknown. Furthermore, some regions of the world are full of riddles, and crossing rooms in these regions takes longer. Fortunately, a leaderboard with the players' best times is provided, starting from a few different rooms. Due to rounding errors, you assume that the best times reported in the leaderboard are measurements affected by additive Gaussian noise $\mathcal{N}(0, 1)$.

i	Starting room	Measured best time
1	$(4, 5)$	2.0
2	$(5, 3)$	3.5
3	$(3, 3)$	4.5
4	$(4, 1)$	7.0
5	$(1, 2)$	8.5

From the leaderboard, you wish to learn a heuristic approximating the best time to get to the exit, starting from room (x, y). You decide to use a small neural network as approximator, described by the following parametric function,

$$h(x, y; \phi) = \mathrm{ReLU}(xw_1 + yw_2 + w_3) + \mathrm{ReLU}(xw_4 + yw_5 + w_6)$$
$$\mathrm{ReLU}(x) = \max(x, 0),$$

where $\phi = (w_1, w_2, w_3, w_4, w_5, w_6)$ is the set of parameters/weights of the neural network.

(a) Among the following sets of parameters (A, B or C), which one would you use? Justify your answer.

Set	w_1	w_2	w_3	w_4	w_5	w_6
ϕ_A	-1.5	1	4	1	-1.5	6
ϕ_B	-1	1.5	3	0	-1	4
ϕ_C	-2	0.5	4.5	1.5	0	5

Solution. A set of parameters is better than another if it explains the data better, that is if the likelihood is higher. Our task is to find the set of parameters that maximizes the likelihood under the assumed probability model. In our case, the data d is the leaderboard and consists of independent position-time tuples (x_i, y_i, t_i). From the statement, we gather that the (best) time t is a function of the starting position (x, y), approximated by a neural network $h(x, y; \phi)$. We also learn that our time measurements are affected by Gaussian

noise.

$$\begin{aligned}
\phi_{\text{MLE}} &= \arg\max_{\phi} p(\mathbf{d} \mid \phi) \\
&= \arg\max_{\phi} \prod_{i=1}^{5} p(x_i, y_i, t_i \mid \phi) \\
&= \arg\max_{\phi} \prod_{i=1}^{5} p(t_i \mid x_i, y_i, \phi)\, p(x_i, y_i) \\
&= \arg\max_{\phi} \prod_{i=1}^{5} \mathcal{N}(t_i; h(x_i, y_i; \phi), 1) \\
&= \arg\max_{\phi} \sum_{i=1}^{5} \log\left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(t_i - h(x_i, y_i; \phi))^2}{2}\right)\right] \\
&= \arg\min_{\phi} \sum_{i=1}^{5} (t_i - h(x_i, y_i; \phi))^2
\end{aligned}$$

Now that we have an objective $L(\phi)$ to minimize, we can evaluate it for each of the proposed sets and select the best one.

$$\begin{aligned}
L(\phi_A) &= \sum_{i=1}^{5} (t_i - h(x_i, y_i; \phi_A))^2 = (2.0 - 5.5)^2 + (3.5 - 6.5)^2 + \cdots = 29.75 \\
L(\phi_B) &= 35.75 \\
L(\phi_C) &= 205.25
\end{aligned}$$

The best set is ϕ_A.

(b) You now assume a Gaussian prior $\mathcal{N}(0, 1)$ on each parameter. Which set of parameters in the table above would you now choose? Justify your answer.

Solution. Now that we hold a prior belief on the parameters, our task is to find the most likely set of parameters given the data, that is maximum a posteriori (MAP) estimation.

$$\begin{aligned}
\phi_{\text{MAP}} &= \arg\max_{\phi} p(\phi \mid \mathbf{d}) \\
&= \arg\max_{\phi} p(\mathbf{d} \mid \phi) p(\phi) \\
&= \arg\max_{\phi} \prod_{i=1}^{5} p(x_i, y_i, t_i \mid \phi) \prod_{j=1}^{6} p(w_j) \\
&= \arg\max_{\phi} \prod_{i=1}^{5} \mathcal{N}(t_i; h(x_i, y_i; \phi), 1) \prod_{j=1}^{6} \mathcal{N}(w_j; 0, 1) \\
&= \arg\min_{\phi} \sum_{i=1}^{5} (t_i - h(x_i, y_i; \phi))^2 + \underbrace{\sum_{j=1}^{6} (w_j)^2}_{\|\phi\|^2}
\end{aligned}$$

Here again, we have an objective $L'(\phi)$ to minimize, which we evaluate for each of the proposed sets.

$$\begin{aligned}
L'(\phi_A) &= L(\phi_A) + \sum_{j=1}^{6} (w_j)^2 = 29.75 + (-1.5)^2 + (1)^2 + \cdots = 88.25 \\
L'(\phi_B) &= 65.0 \\
L'(\phi_C) &= 257.0
\end{aligned}$$

The best set is ϕ_B.

(c) Discuss the procedure you would implement on a computer to find the optimal set of parameters, had the table above not been provided.

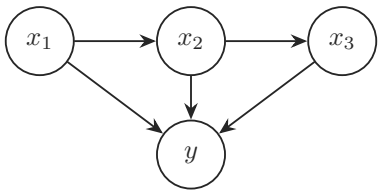
Solution. If the table is not provided, it is intractable to evaluate and compare all possible sets of parameters. However, we still want to minimize the MLE (or MAP) objective $L(\phi)$. We can improve a set of parameters ϕ by following the opposite of the objective's gradient $\nabla_\phi L(\phi)$, *i.e.* performing *gradient descent* steps

$$\phi \leftarrow \phi - \alpha \nabla_\phi L(\phi),$$

where α is the learning rate.

Exercise 2 Predict the leaderboard score

A leaderboard ranks language models by an overall score. The Bayesian network below represents how this score is computed. In this model, x_1, x_2 and x_3 respectively denote the scores obtained by a model on a validation set, on a public benchmark and on a held-out test set. The three evaluations share the same harness, and a model that does well on one tends to do well on the next. In particular, $x_2 \sim \mathcal{N}(a_1 x_1 + \mu_2, \sigma_2^2)$ and $x_3 \sim \mathcal{N}(a_2 x_2 + \mu_3, \sigma_3^2)$. Finally, $y \sim \mathcal{N}(a_3 x_1 + a_4 x_2 + a_5 x_3 + \mu_y, \sigma_y^2)$ stands for the overall score, which is a linear combination of the three scores plus some Gaussian noise due to rounding errors. Answer the following questions about this model.



(a) Assuming the parameters of the model are known, what is the expected value of y given x_1 and x_2?

Solution. Our task is to find the expectation

$$\mathop{\mathbb{E}}_{p(y|x_1,x_2)}[y] = \int y\, p(y \mid x_1, x_2)\, \mathrm{d}y.$$

We know that

$$p(y \mid x_1, x_2) = \int p(y \mid x_1, x_2, x_3)\, p(x_3 \mid x_1, x_2)\, \mathrm{d}x_3,$$

where $p(y \mid x_1, x_2, x_3)$ is given in the statement and $p(x_3 \mid x_1, x_2) = p(x_3 \mid x_2)$, since $x_3 \perp x_1 \mid x_2$. Both are linear Gaussian distributions. Therefore, we have

$$p(y \mid x_1, x_2) = \mathcal{N}\Big(y \mid a_3 x_1 + a_4 x_2 + a_5(a_2 x_2 + \mu_3) + \mu_y, (a_5 \sigma_3)^2 + \sigma_y^2\Big)$$

and, by definition of a Gaussian distribution,

$$\mathop{\mathbb{E}}_{p(y|x_1,x_2)}[y] = a_3 x_1 + (a_4 + a_5 a_2) x_2 + a_5 \mu_3 + \mu_y.$$

(b) Suppose now that the parameters are unknown, and that the scores x_3 on the held-out test set are not observed. Given a learning set $\mathbf{d} = \{(x_{i,1}, x_{i,2}, y_i)\}$ of N independent and identically distributed points, derive the maximum likelihood estimate of the parameters of $p(y \mid x_1, x_2)$.

Solution. We know that the distribution of y given x_1 and x_2 takes the form $\mathcal{N}(y \mid w_1 x_1 + w_2 x_2 + b, \sigma^2)$. Then, our task is to find the parameters $h = (w_1, w_2, b, \sigma)$ that maximize the likelihood of d, *i.e.* the *maximum likelihood estimation* (MLE)

$$\begin{aligned}
h_{\text{MLE}} &= \arg\max_h p(\mathbf{d} \mid h) \\
&= \arg\max_h \prod_i p(x_i, y_i \mid h) \\
&= \arg\max_h \log \prod_i p(x_i, y_i \mid h) \\
&= \arg\max_h \sum_i \log p(x_i, y_i \mid h) \\
&= \arg\max_h \sum_i \log p(y_i \mid h, x_i) + \log p(x_i) \\
&= \arg\max_h \sum_i \log p(y_i \mid h, x_i) \\
&= \arg\max_h \sum_i \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(w^T x_i - y_i)^2}{2\sigma^2} \right) \right] \\
&= \arg\max_h \sum_i -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(w^T x_i - y_i)^2}{2\sigma^2} \\
&= \arg\min_h \log\sigma^2 + \frac{1}{\sigma^2}\frac{1}{N}\sum_i (w^T x_i - y_i)^2,
\end{aligned}$$

where $x_i = \begin{pmatrix} x_{i,1} & x_{i,2} & 1 \end{pmatrix}^T$ and $w = \begin{pmatrix} w_1 & w_2 & b \end{pmatrix}^T$. In the last expression, we observe that the minimizer of the summation term is independent of σ. Therefore,

$$w_{\text{MLE}} = \arg\min_w \sum_i (w^T x_i - y_i)^2,$$

which exactly corresponds to a *linear regression* problem. Then, we find w_{MLE} by canceling the gradient with respect to w, *i.e.*

$$\begin{aligned}
0 &= \nabla_w \sum_i (w^T x_i - y_i)^2 \\
&= \nabla_w \sum_i (w^T x_i - y_i)(w^T x_i - y_i) \\
&= \nabla_w \sum_i \left((w^T x_i)^2 + y_i^2 - 2 w^T x_i y_i \right) \\
&= \nabla_w \left(w^T X^T X w + Y^T Y - 2 w^T X^T Y \right) \\
&= 2 X^T X w - 2 X^T Y
\end{aligned}$$

where $X = (x_i^T) \in \mathbb{R}^{N \times 3}$ and $Y = (y_i) \in \mathbb{R}^N$. Finally, we have

$$\begin{aligned}
0 &= X^T X w_{\text{MLE}} - X^T Y \\
\Leftrightarrow \quad w_{\text{MLE}} &= (X^T X)^{-1} X^T Y.
\end{aligned}$$

Afterwards, we find σ_{MLE} such that

$$\sigma_{\text{MLE}} = \arg\min_\sigma \log\sigma^2 + \frac{\text{MSE}}{\sigma^2}$$

$$= \sqrt{\text{MSE}},$$

where MSE denotes the *mean squared error*

$$\frac{1}{N}\sum_i ({w_{\text{MLE}}}^T x_i - y_i)^2.$$

By (a), the fitted parameters estimate combinations of those of the network, $w_1 = a_3$, $w_2 = a_4 + a_5 a_2$, $b = a_5\mu_3 + \mu_y$ and $\sigma^2 = a_5^2\sigma_3^2 + \sigma_y^2$. Without observations of x_3, the individual parameters $a_2, a_4, a_5, \mu_3, \mu_y, \sigma_3, \sigma_y$ cannot be recovered separately.

Exercise 3 Heteroscedastic linear regression

In linear regression with Gaussian noise of constant variance, $y_i \sim \mathcal{N}(w^T x_i, \sigma^2)$, the maximum likelihood estimate of the weights is $w_{\text{MLE}} = (X^T X)^{-1} X^T Y$, where the rows of X are the inputs x_i^T and $Y = (y_i)$. What does this estimate become if the noise is different for each sample, *i.e.* $y_i \sim \mathcal{N}(w^T x_i, \sigma_i^2)$ with known variances σ_i^2?

Solution. The log-likelihood of the weights is

$$L(w) = \sum_i \log \mathcal{N}(y_i \mid w^T x_i, \sigma_i^2) = -\sum_i \frac{(w^T x_i - y_i)^2}{2\sigma_i^2} + \text{const},$$

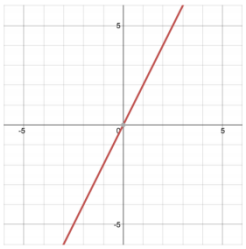
so that maximizing it minimizes a weighted sum of squares, each sample weighted by its precision $1/\sigma_i^2$. With $W = \text{diag}(1/\sigma_1^2, \ldots, 1/\sigma_N^2)$,

$$\nabla_w L(w) = -X^T W(Xw - Y) = 0 \quad \Leftrightarrow \quad w_{\text{MLE}} = (X^T W X)^{-1} X^T W Y.$$

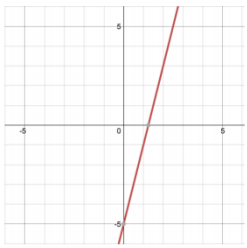
Noisy samples count less. With equal variances, W is proportional to the identity and the estimate of the homoscedastic case is recovered.

Exercise 4 Neural network representations (CS188, Spring 2024)

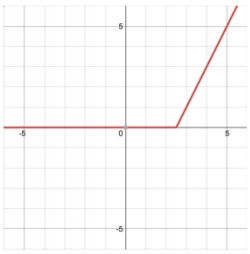
You are given a number of functions (a-h) of a single variable, x, which are graphed below. For each network structure proposed afterwards, indicate which functions they are able to represent. When possible, indicate appropriate values for the parameters.



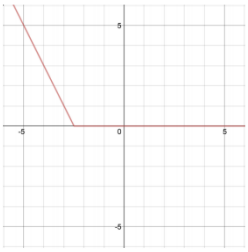
(a) $2x$



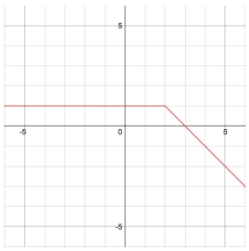
(b) $4x - 5$



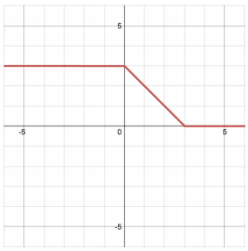
(c) $\begin{cases} 2x - 5 & x \geq 2.5 \\ 0 & x < 2.5 \end{cases}$



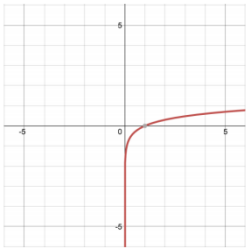
(d) $\begin{cases} -2x - 5 & x \leq -2.5 \\ 0 & x > -2.5 \end{cases}$



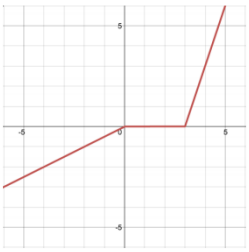
(e) $\begin{cases} -x + 3 & x \geq 2 \\ 1 & x < 2 \end{cases}$



(f) $\begin{cases} 3 & x \leq 0 \\ 3 - x & 0 < x \leq 3 \\ 0 & x > 3 \end{cases}$

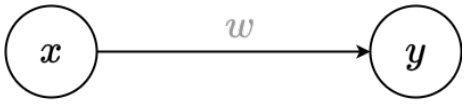


(g) $\log(x)$



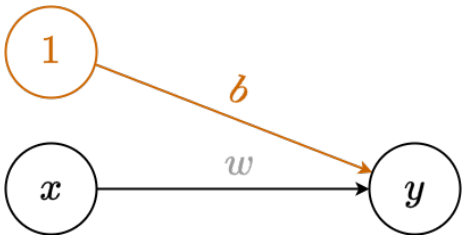
(h) $\begin{cases} 0.5x & x \leq 0 \\ 0 & 0 < x \leq 3 \\ 3x - 9 & x > 3 \end{cases}$

(a) We start with a linear transformation of the scalar input x, weight w, and output y, such that $y = wx$.



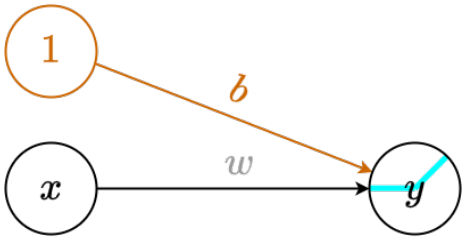
Solution. This network can only represent (a), with $w = 2$. Since there is no bias term, the line must pass through the origin.

(b) We then introduce a bias term b, such that $y = wx + b$.



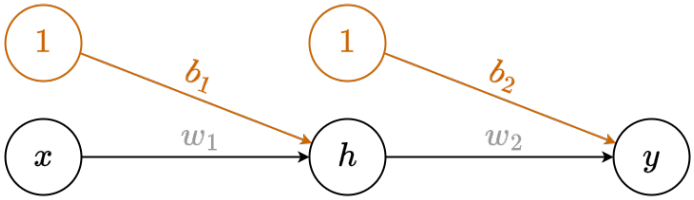
Solution. (a) with $w = 2$ and $b = 0$, and (b) with $w = 4$ and $b = -5$.

(c) We now introduce a non-linearity $\sigma(.)$ into the network. We use the ReLU non-linearity, which has the form $\text{ReLU}(x) = \max(0, x)$.



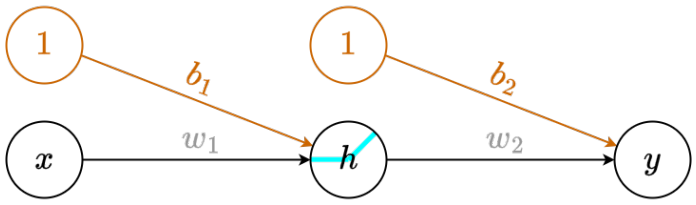
Solution. With the output coming directly from the ReLU, this cannot produce any values less than zero. It can produce (c) with $w = 2$ and $b = -5$, and (d) with $w = -2$ and $b = -5$.

(d) Now we consider neural networks with multiple affine transformations, as depicted below.



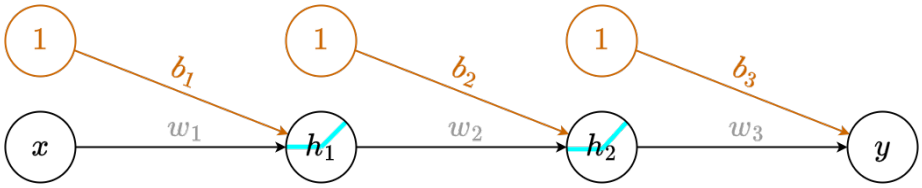
Solution. Applying multiple affine transformations (with no non-linearity in between) is not any more powerful than a single affine function: $w_2(w_1x + b_1) + b_2 = w_2w_1x + w_2b_1 + b_2$, so this is just an affine function with different coefficients. The functions we can represent are the same as in (b), if we choose $w_1 = w$, $w_2 = 1$, $b_1 = 0$, $b_2 = b$: (a) with $w_1 = 2$, $w_2 = 1$, $b_1 = 0$, $b_2 = 0$, and (b) with $w_1 = 4$, $w_2 = 1$, $b_1 = 0$, $b_2 = -5$.

(e) We now add a ReLU non-linearity to the network between the affine transformations.



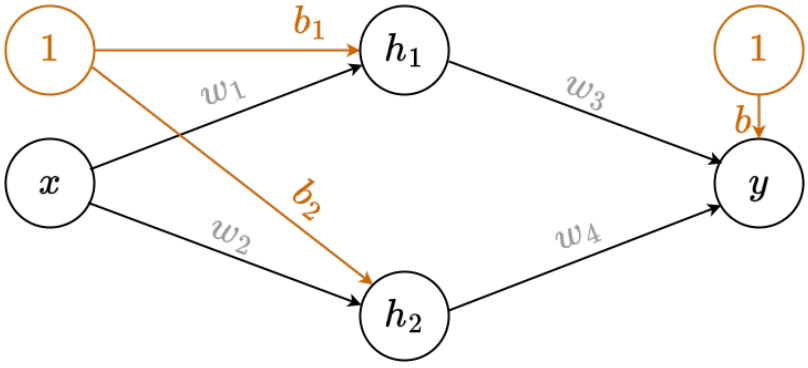
Solution. (c), (d), and (e). The affine transformation after the ReLU is capable of stretching (or flipping) and shifting the ReLU output in the vertical dimension. The parameters to produce these are: (c) with $w_1 = 2$, $b_1 = -5$, $w_2 = 1$, $b_2 = 0$, (d) with $w_1 = -2$, $b_1 = -5$, $w_2 = 1$, $b_2 = 0$, and (e) with $w_1 = 1$, $b_1 = -2$, $w_2 = -1$, $b_2 = 1$.

(f) Now we add another hidden layer to the network, as depicted below. You do not have to guess parameters values anymore.



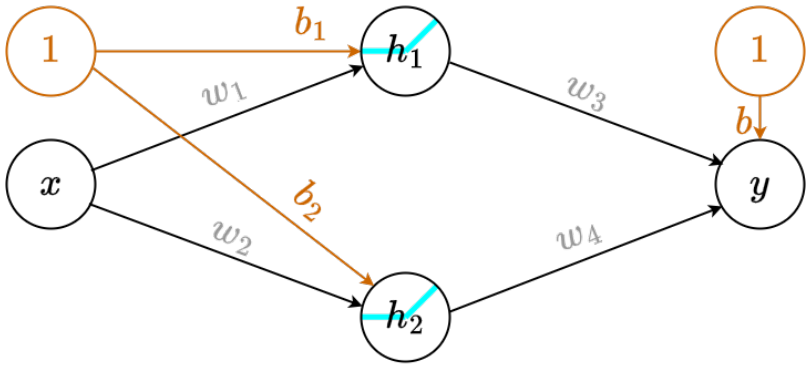
Solution. (c), (d), (e), and (f). The network can represent all the same functions as (e) (because note that we could have $w_2 = 1$ and $b_2 = 0$). In addition it can represent (f): the first ReLU can produce the first flat segment, the affine transformation can flip and shift the resulting curve, and then the second ReLU can produce the second flat segment (with the final affine layer not doing anything). Note that (h) cannot be produced since its line has only one flat segment (and the affine layers can only scale, shift, and flip the graph in the vertical dimension; they can't rotate the graph).

(g) We'd like to consider using a neural net with just one hidden layer, but larger. Let's first consider using just two affine functions, with no nonlinearity in between. You do not have to guess parameters values anymore.



Solution. (a) and (b). With no non-linearity, this reduces to a single affine function (in the same way as (d)).

(h) Now we'll add a non-linearity between the two affine layers, to produce the neural network below with a hidden layer of size 2. You do not have to guess parameters values anymore.



Solution. All functions except for (g). Note that we can recreate any network of (e) by setting w_4 to 0, so this allows us to produce (c), (d) and (e). To produce the rest of the functions, note that h_1 and h_2 will be two independent functions with a flat part lying on the x-axis, and a portion with positive slope. The final layer takes a weighted sum of these two functions. To produce (a) and (b), the flat portion of one ReLU should start at the point where the other ends ($x = 0$ for (a), or $x = 1$ for (b)). The final layer then vertically flips the ReLU sloping down and adds it to the one sloping up, producing a single sloped line. To produce (h), the ReLU sloping down should have its flat portion end (at $x = 0$) before the other's flat portion begins (at $x = 3$). The down-sloping one is again flipped and added to the up-sloping. To produce (f), both ReLUs should have equal slope, which will cancel to produce the first flat portion above the x-axis.

(i) Are there functions that can't be represented by all proposed networks? If yes, explain why and what you would need to model them.

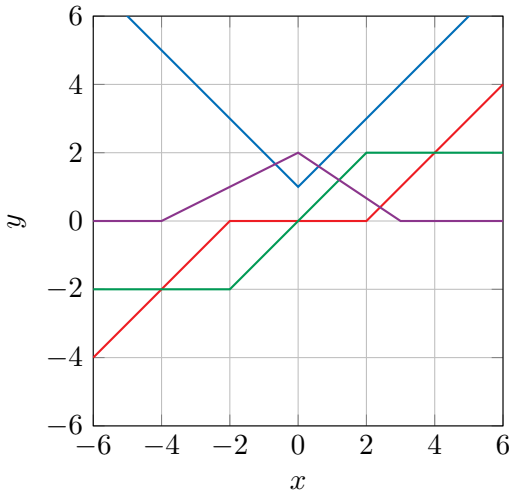
Solution. Yes, function (g) cannot be modeled. We first see that the function is smooth and representing it only with few ReLU activations is not enough. We should probably use smoother activations like SiLU, tanh, ... (see the activation functions of the "Machine learning and neural networks" lecture).

However, we remember the universal approximation theorem which states that a single-hidden-layer network with a finite number of hidden units can approximate any continuous function on a compact subset of $\mathbb{R}^d$ to arbitrary accuracy. Then, adding sufficient amount of hidden units of the 1-layer network of (h) should be sufficient.

Note that the logarithm function is only defined on $\mathbb{R}^+$. If our network is made of activation functions defined on $\mathbb{R}$, we will never be able to restrict the domain of the learned function to only real positive. However, you can be as accurate as you want on the desired domain, but it makes no sense to evaluate your network elsewhere for this problem.

Exercise 5 Piecewise-linear functions

For each of the piecewise-linear functions below, write a function $y = f(x)$ as a composition of sums $(+, -)$, ReLU ($\text{ReLU}(x) = \max(x, 0)$) non-linearities, and real parameters (weights and biases) that exactly matches the function over $\mathbb{R}$.



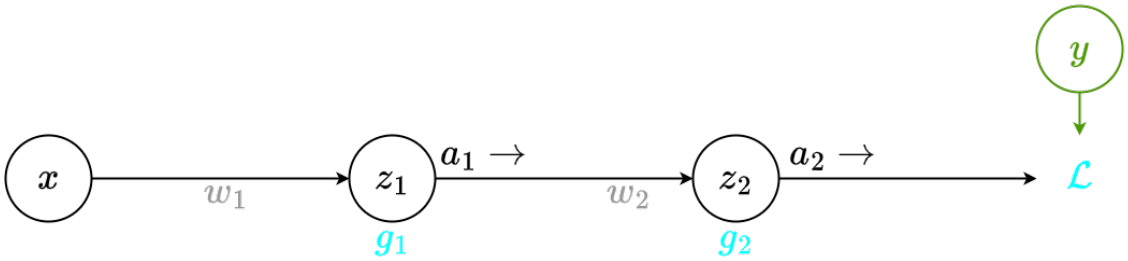
For example, $y = \text{ReLU}(x + 2) - \text{ReLU}(-2x)$ is a valid function.

Solution.

1. $y = \text{ReLU}(x) - \text{ReLU}(-x) + 1$
2. $y = \text{ReLU}(x - 2) - \text{ReLU}(-x - 2)$
3. $y = \text{ReLU}(-\text{ReLU}(-x + 2) + 4) - 2$
4. $y = \text{ReLU}(-\text{ReLU}(\frac{2}{3}x) - \text{ReLU}(-\frac{1}{2}x) + 2)$

Exercise 6 Binary classification (CS188, Spring 2024)

Consider the following simple neural network for binary classification. Here x is a single real-valued input feature with an associated class y (0 or 1). There are two weight parameters w_1 and w_2, and non-linearity functions g_1 and g_2. The network will output a value a_2 between 0 and 1, representing the probability of being in class 1. We will be using a loss function $\mathcal{L}$ to compare the prediction a_2 with the true class y.



(a) Write down the forward pass on this network, writing the output values for each node z_1, a_1, z_2 and a_2 in terms of the node's input values.

Solution.

$$z_1 = w_1 x$$
$$a_1 = g_1(z_1) = g_1(w_1 x)$$
$$z_2 = w_2 a_1 = w_2 g_1(w_1 x)$$
$$a_2 = g_2(z_2) = g_2(w_2 g_1(w_1 x))$$

(b) Derive the arguments of the loss $\mathcal{L}(a_2, y)$ in terms of the input x, weights w_i, and activation functions g_i.

Solution. The argument a_2 has been derived in the previous question. We have

$$a_2 = g_2(w_2 g_1(w_1 x)).$$

For the other input y, the latter is given as input of the problem. Indeed, in supervised learning, we assume access to a training dataset made of input-output pairs $\{(x_i, y_i)\}^N$ where y_i acts as a target for the network.

(c) Using the chain rule, differentiate $\mathcal{L}$ w.r.t. w_2. Write your expression as a product of partial derivatives at each node: *i.e.* the partial derivative of the node's output with respect to its inputs. (Hint: the series of expressions you wrote in (a) will be helpful; you may use any of those variables.)

Solution.

$$\frac{\partial \mathcal{L}}{\partial w_2} = \frac{\partial \mathcal{L}}{\partial a_2} \frac{\partial a_2}{\partial z_2} \frac{\partial z_2}{\partial w_2}$$

(d) Motivate a choice for the output activation function g_2 given the nature of the problem. Motivate also a loss function. Give the expression of the gradients of both functions w.r.t. to their inputs.

Solution. From the statement

(a) we know that we are in a classification problem (specifically, a binary classification),

(b) we know that the output a_2 should be between 0 and 1 and represents the probability $p(Y = 1 \mid x)$.

From (b), we have $a_2 : \mathbb{R} \mapsto [0, 1]$. A typical choice that fulfill this requirement is the sigmoid function $\sigma(z) = \dfrac{1}{1 + e^{-z}}$. The gradient of the sigmoid w.r.t. its input is

$$\frac{\partial \sigma(z)}{\partial z} = \frac{e^{-z}}{(1 + e^{-z})^2} = \sigma(z)\sigma(-z) = \sigma(z)(1 - \sigma(z)).$$

For the loss function, either we remember the cross-entropy loss seen in the theoretical lecture, or we compute it from maximum likelihood estimation.
The likelihood of a pair (x, y) can be derived from our model as

$$\begin{cases} \text{if } y = 1 \quad \text{then} \quad P(Y = y \mid x) = \mathrm{NN}_\theta(x) \\ \text{if } y = 0 \quad \text{then} \quad P(Y = y \mid x) = 1 - \mathrm{NN}_\theta(x) \end{cases}$$

where NN_θ is the neural network parameterized by $\theta = [w_1, w_2]$. Denoting $\mathrm{NN}_\theta(x) = a_2$, the likelihood expression for one pair can be reformulated as

$$L(x, y; \theta) = y \log a_2 + (1 - y) \log(1 - a_2).$$

Since we want to maximize the likelihood, but minimize the loss, we can set the latter to

$$\mathcal{L}(a_2, y) = -y \log a_2 - (1 - y) \log(1 - a_2)$$

which corresponds to the binary cross-entropy loss. The gradient of the loss w.r.t. a_2 is

$$\frac{\partial \mathcal{L}(a_2, y)}{\partial a_2} = -\frac{y}{a_2} + \frac{(1 - y)}{(1 - a_2)} = \frac{a_2 - y}{a_2(1 - a_2)}.$$

(e) We set the loss function to the binary cross-entropy

$$\mathcal{L}(a_2, y) = -y \log a_2 - (1-y)\log(1-a_2),$$

and g_1 and g_2 are both sigmoid functions $\sigma(z) = \frac{1}{1+e^{-z}}$. If you made this choice in the previous question, use your results to get the expression of $\frac{\partial \mathcal{L}}{\partial w_2}$. Otherwise, start by expressing $\frac{\partial \sigma(z)}{\partial z}$ and $\frac{\partial \mathcal{L}}{\partial a_2}$.

Solution. We first miss $\frac{\partial z_2}{\partial w_2}$ which is

$$\frac{\partial z_2}{\partial w_2} = \frac{\partial}{\partial w_2}(w_2 a_1) = a_1.$$

Plugging all the partial derivatives into the expression found in (c), we have

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial w_2} &= \frac{a_2 - y}{a_2(1-a_2)} g_2(z_2)(1-g_2(z_2))a_1 \\ &= \frac{a_2 - y}{a_2(1-a_2)} a_2(1-a_2)a_1 \\ &= (a_2 - y)a_1\end{aligned}$$

(f) Now use the chain rule to express $\frac{\partial \mathcal{L}}{\partial w_1}$ as a product of partial derivatives at each node of interest.

Solution.

$$\frac{\partial \mathcal{L}}{\partial w_1} = \frac{\partial \mathcal{L}}{\partial a_2}\frac{\partial a_2}{\partial z_2}\frac{\partial z_2}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_1}$$

(g) Finally, write $\dfrac{\partial \mathcal{L}}{\partial w_1}$ in terms of x, y, w_i, a_i, z_i.

Solution. We first need to compute

$$\frac{\partial a_1}{\partial z_1} = g_1(z_1)(1-g_1(z_1)) = a_1(1-a_1)$$

similarly to $\frac{\partial a_2}{\partial z_2}$, and

$$\frac{\partial z_1}{\partial w_1} = \frac{\partial}{\partial w_1}(w_1 x) = x.$$

Then, we get

$$\frac{\partial \mathcal{L}}{\partial w_1} = (a_2 - y)w_2 a_1(1-a_1)x.$$

(h) What is the gradient descent update for w_1 with learning rate α in terms of the values computed above?

Solution.

$$w_1 \leftarrow w_1 - \alpha(a_2 - y)w_2 a_1(1-a_1)x.$$

You see here the importance of the forward pass. Iteratively, when receiving the input x, we compute all the intermediate activations (here, a_1 and a_2) and the loss (given y). Then, we propagate backward (from the output to the input) the values following the chain rule for the gradient of the loss w.r.t. parameters of interest. Finally, we update the parameters using the computed gradient.

Exercise 7 Training loop (Exam, January 2024)

Let us consider a neural network f with one hidden layer taking as input a scalar $x \in \mathbb{R}$ and producing as output a scalar $y = f(x;\theta) \in \mathbb{R}$. The neural network is defined as

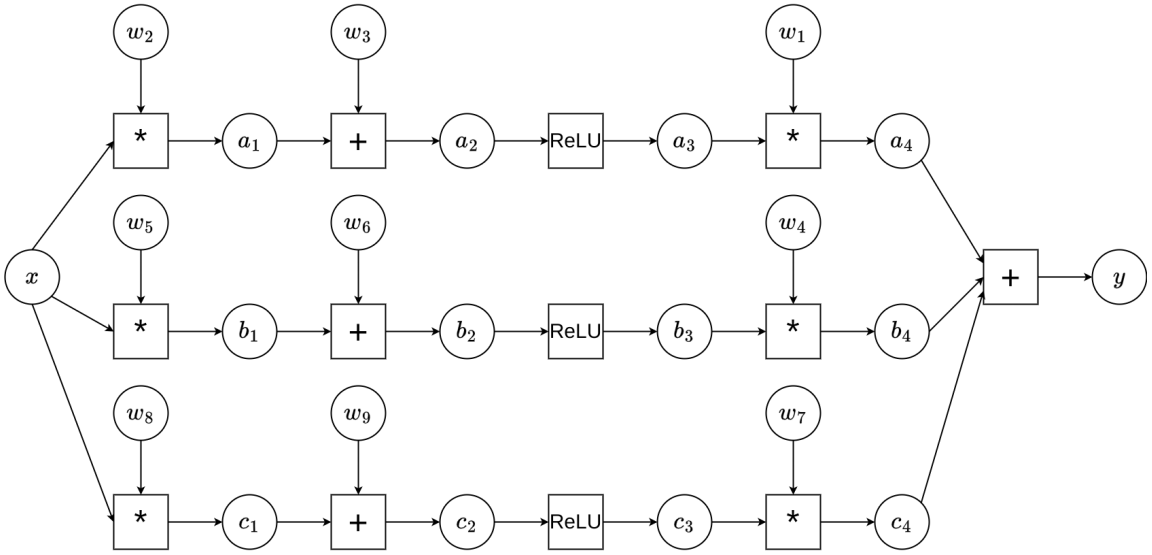
$$f(x;\theta) = w_1 \text{ReLU}(w_2 x + w_3) + w_4 \text{ReLU}(w_5 x + w_6) + w_7 \text{ReLU}(w_8 x + w_9),$$

where $\theta = (w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8, w_9)$ is the set of parameters and $\text{ReLU}(x) = \max(x, 0)$ is the rectified linear unit function.

(a) Draw the computation graph representing the neural network and the flow of information from inputs to outputs. Your diagram should be a directed graph that follows the following conventions:

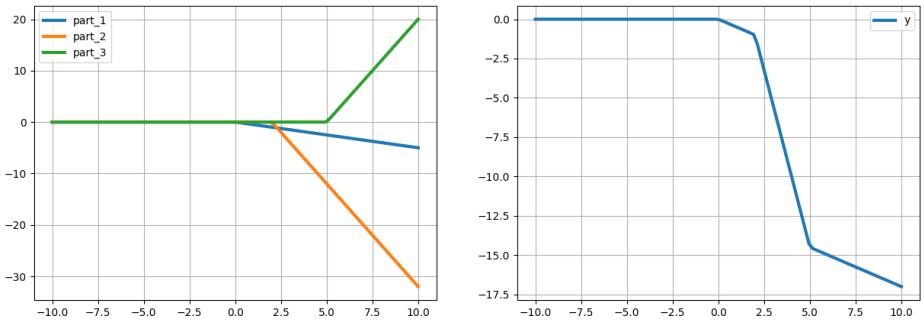
- circled nodes correspond to variables (input, output, parameters or intermediate variables),
- squared nodes correspond to primitive operations (addition, multiplication, ReLU) and produce an intermediate variable as output,
- directed edges correspond to the flow of information, from inputs to outputs.

Solution.



(b) For $\theta = (-1, \frac{1}{2}, 0, -4, 1, -2, 4, 1, -5)$, draw the function $y = f(x;\theta)$ for $x \in [-10, 10]$.

Solution.



(c) Using the data point $(x, y) = (10, -15)$ and the value of θ given above, we want to fine-tune the parameter w_8 of the neural network such that $f(x; \theta)$ produces a more accurate prediction of y.

i. Evaluate the squared error loss at the data point $(x, y) = (10, -15)$ and the current value of θ.
Solution.

$$f(10, \theta) = -17$$
$$\mathcal{L}(\hat{y} = -17, y = -15) = (17 - 15)^2 = 4$$

ii. Derive an expression for the derivative of the squared error loss with respect to w_8.
Solution. The loss is

$$\mathcal{L}(\hat{y}, y) = (\hat{y} - y)^2.$$

We compute the gradient

$$\frac{\partial \mathcal{L}}{\partial w_8} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial c_4} \frac{\partial c_4}{\partial c_3} \frac{\partial c_3}{\partial c_2} \frac{\partial c_2}{\partial c_1} \frac{\partial c_1}{\partial w_8}$$
$$= 2(\hat{y} - y) w_7 \frac{\partial c_3}{\partial c_2} x.$$

We need to compute forward activations to compute the gradient value. $\frac{\partial c_3}{\partial c_2}$ will be 1 if $c_2 > 0$ else 0 by definition of the ReLU. We have

$$\hat{y} = -17$$
$$y = -15$$
$$w_7 = 4$$
$$x = 10$$
$$c_2 = w_8 x + w_9 = 5 \Rightarrow \frac{\partial c_3}{\partial c_2} = 1$$

which gives a gradient equal to -160.

iii. Update the parameter w_8 using one step of gradient descent with a learning rate $\alpha = 0.0005$.
Solution. The gradient step is given by

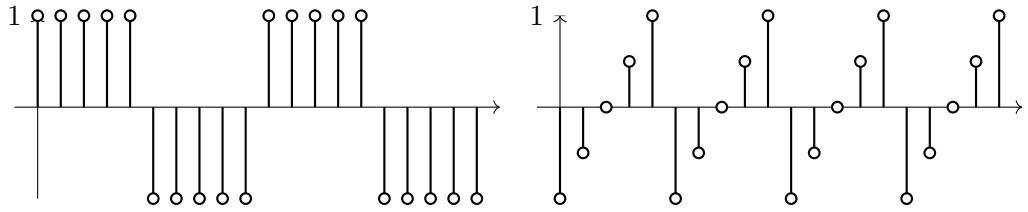
$$w_8 \leftarrow w_8 - \alpha \frac{\partial \mathcal{L}}{\partial w_8}$$

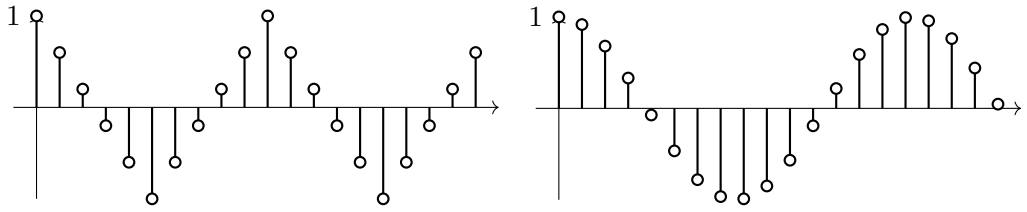
which gives an updated w_8 equal to 1.08.

iv. Verify if the value of the loss function has decreased after the update.
Solution. With the updated w_8, we have $\hat{y} = -13.8$ which gives a loss equal to 1.44. The previous loss value was 4, we indeed decreased the loss.

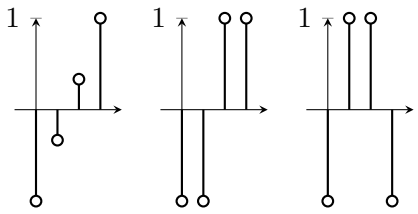
Exercise 8 1d convolution

Represent the convolution $x \circledast \omega$ of each of the following signals x

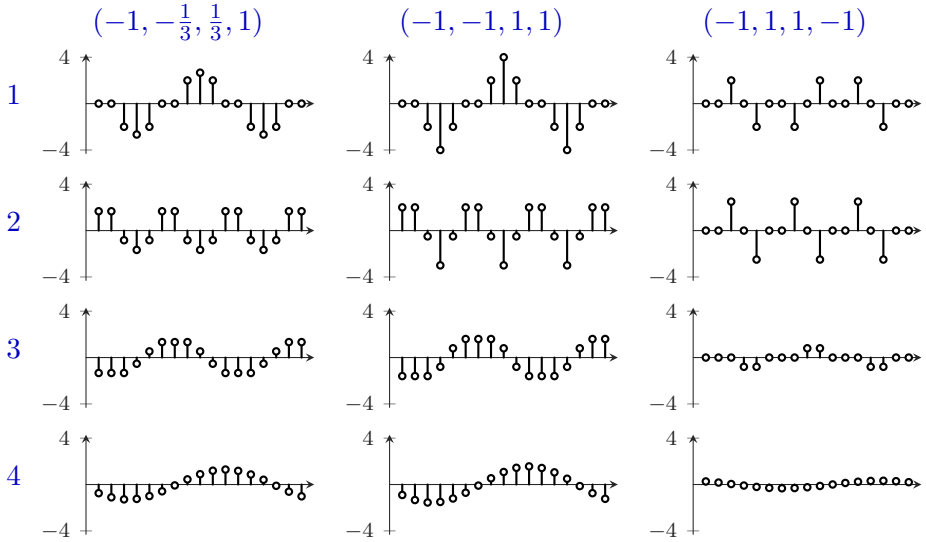




with each of the following convolution kernels ω.



Solution. Without padding, each output has $20 - 4 + 1 = 17$ values, $(x \circledast \omega)[i] = \sum_{m=0}^{3} x_{i+m}\omega_m$. The kernels are $\omega = (-1, -\frac{1}{3}, \frac{1}{3}, 1)$, $(-1, -1, 1, 1)$ and $(-1, 1, 1, -1)$; the outputs below share the vertical scale $[-4, 4]$, one row per signal and one column per kernel.



The first kernel is a smoothed derivative: its output is positive where the signal increases, negative where it decreases and zero where it is constant, so that the triangle wave gives a rounded square wave and the cosine a sine. The second compares the two halves of its window: it also measures the slope, but responds most strongly to jumps, with peaks of ± 4 at the edges of the square wave. The third compares the centre of its window with its ends: it is zero wherever the signal is linear (the flat parts of the square wave, the ramps of the sawtooth and of the triangle), positive at local maxima and negative at local minima, so that it detects corners; on the cosine, whose curvature is proportional to the signal itself, it returns a small copy of the signal.

Exercise 9 2d convolution

(a) Represent the 2D convolution $x \circledast \omega$ of the following greyscale image x, where a value of 0 corresponds to a white pixel and a value of 1 corresponds to a black pixel,

0.1	0.3	0.6	0.9	0.4
0.2	0.5	0.8	0.7	0.3
0.4	0.7	1.0	0.8	0.5
0.2	0.6	0.9	0.7	0.3
0.1	0.4	0.7	0.5	0.2

with the following convolution kernel ω

0.2	0.6	0.9
0.4	0.8	0.3

Is the result dependent on the choice of coordinate system (*i.e.* on which directions are considered the "i" and "j" directions in $x[j, i]$ and $\omega[j, i]$)?

Solution. The result is equivariant to the choice of coordinate system. The resulting feature map, as drawn on the sheet of paper, remains the same. Only the interpretation of which pixel is represented by $(x \circledast \omega)[j, i]$ changes.

1.46	2.28	1.99
2.08	2.53	2.04
2.23	2.63	2.14
1.78	2.16	1.61

(b) The image x now is a coloured image and has three channels (one for the red colour, one for the green colour and one for the blue colour):

$x[0, :, :]$

0.1	0.3	0.6	0.9
0.2	0.5	0.8	0.7
0.4	0.7	1.0	0.8
0.2	0.6	0.9	0.7

$x[1, :, :]$

0.2	0.2	0.2	1.0
0.2	0.2	0.2	1.0
0.2	0.2	0.2	1.0
0.2	0.2	0.2	1.0

$x[2, :, :]$

0.2	0.2	0.2	0.1
0.2	0.2	0.2	0.1
0.2	0.2	0.2	0.1
0.2	0.2	0.2	0.1

Represent the 2D convolution $x \circledast \omega$ with the following convolution kernel ω

$\omega[0, :, :]$

0.2	0.6	0.9
0.4	0.8	0.3

$\omega[1, :, :]$

0.1	0.1	0.1
0.1	0.1	0.1

$\omega[2, :, :]$

0.2	0.2	0.2
0.2	0.2	0.2

Is the result dependent on the choice of coordinate system? What happens if the channels of x are shuffled?

Solution. The dimension corresponding to the channels cannot be interchanged with those corresponding to the height or width of the image while preserving the validity and correctness of the operation. This result makes sense, since the operation is a 2D convolution applied to a three-dimensional tensor. Changing the order of the channels in x will also lead to a different convolution result. For the 2D coordinate system used to describe the spatial dimensions of the images at each channel (width and height), the same result as in the previous point holds.

1.82	2.76
2.44	3.01
2.59	3.11

Exercise 10 Problem statement, output layer and loss function

A dataset $\mathbf{d}$ consists of $N = 500$ images of cats and dogs, with exactly one cat or one dog appearing in each image. Each image has a spatial resolution of 32×32 pixels and is encoded in RGB format using three channels of real-valued entries. The first channel represents the red component of the image, the second channel represents the green component, and the third channel represents the blue component.

You would like to formulate a supervised learning problem in order to develop a deep learning model suited to a given task.

You are interested in three different tasks:

1. Assign a beauty score (a real-valued quantity) to a given image.
2. Whether the image contains a cat or a dog.
3. Whether each pixel of an image belongs to a cat, a dog or the background.

For each problem, you want to identify a probabilistic model in the form $p_\theta(y \mid x)$ from a dataset that you will construct:

$$\mathbf{d} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)\}.$$

(a) For each task, define the random variables X and Y, specify their respective dimensions, and explain what θ represents.

Solution. For each task, θ represents the parameters of the model being developed. X is the distribution of the image given as input to the model ($X \in \mathbb{R}^{3,32,32}$).

For the first task, Y is the distribution of the beauty score associated with an image. $Y \in \mathbb{R}$.

For the second task, Y is a binary random variable indicating whether the image contains a dog (let's say $Y = 1$ in this case) or a cat ($Y = 0$): $Y \in \{0, 1\}$.

For the third case, Y is a collection of categorical random variables $P_{u,v}$, each one associated with each pixel of the image X:

$$Y = \bigcup_{v=0}^{31} \bigcup_{u=0}^{31} P_{u,v}.$$

Each $P_{u,v}$ indicates whether pixel (u, v) belongs to a cat, a dog, or the background: $P_{u,v} \in \{\text{cat}, \text{dog}, \text{background}\}$.

Once the task has been formulated, you construct the dataset $\mathbf{d}$ by annotating each of the 500 images, thereby obtaining a sample $y_i \sim Y$ associated with a sampled image $x_i \sim X$.

(b) Show that finding the parameters θ that maximize the likelihood of your model over a sample of $\mathbf{d}$,

$$p\left((x_i, y_i) \mid \theta\right),$$

is equivalent to finding the parameters θ that maximize the probability of the modelled conditional distribution over this sample,

$$p\left(y_i \mid x_i, \theta\right).$$

Solution.

$$\begin{aligned}
\arg\max_\theta \left\{p\left((x_i, y_i) \mid \theta\right)\right\} &= \arg\max_\theta \left\{p\left(y_i \mid x_i, \theta\right) p(x_i \mid \theta)\right\} \\
&= \arg\max_\theta \left\{p\left(y_i \mid x_i, \theta\right) p(x_i)\right\} \\
&\qquad \text{(As the model is over } Y \mid X\text{, not } X\text{.)} \\
&= \arg\max_\theta \left\{p\left(y_i \mid x_i, \theta\right)\right\}
\end{aligned}$$

After reviewing the literature, you choose a model for $p_\theta(y \mid x) = p(y \mid x, \theta)$.
For the first task, you assume that a suitable model is to model $Y \mid X$ as a Gaussian distribution with homoscedastic variance. For the second task, a suitable model is a Bernoulli distribution. For the third task, a suitable model is a categorical distribution.

(c) For each task, propose a suitable loss function to train your neural network. Propose also an output layer for your neural network in the form

$$y = \sigma\left(W_y h + b_y\right).$$

Let h denote the output of the layer immediately preceding the output layer, with dimension d_o ($h \in \mathbb{R}^{d_o}$). Specify the dimensions of y, W_y, and b_y and give an example of a suitable non-linear activation function σ for each task.

Hint: For each task, which parameter of the distribution $p_\theta(y \mid x)$ is estimated by your neural network?

Solution. For the first task, the statistical model is:

$$\begin{aligned}
Y \mid X &\sim \mathcal{N}\left(\mu(X), \sigma^2\right) \\
\rightarrow p(y \mid x) &\propto \exp\left(-\frac{1}{2}\frac{(y - \mu(x))^2}{\sigma^2}\right)
\end{aligned}$$

The neural network estimates the mean of the Gaussian distribution and outputs $\hat{\mu}(X; \theta)$.
For the second task:

$$\begin{aligned}
Y \mid X &\sim \text{Bern}\left(q(X)\right) \\
\rightarrow p(y \mid x) &\propto q(x)^y (1 - q(x))^{(1-y)}
\end{aligned}$$

The neural network estimates the probability that the image contains a dog and outputs $\hat{q}(X; \theta)$.
For the third task:

$$\begin{aligned}
Y \mid X &\sim \text{Categ}\left(q_1(X), q_2(X), q_3(X)\right) \\
\rightarrow p(y \mid x) &\propto (q_1(x))^{\mathbb{I}(y=\text{dog})} (q_2(x))^{\mathbb{I}(y=\text{cat})} (q_3(x))^{\mathbb{I}(y=\text{background})}
\end{aligned}$$

with $q_1(x) + q_2(x) + q_3(x) = 1$ and with $\mathbb{I}(y = c)$ being 1 if the pixel y belongs to the class c, 0 otherwise. The neural network estimates the probability that each pixel belongs to a dog, a cat, or the background. It outputs $\hat{q}_1(X;\theta)$, $\hat{q}_2(X;\theta)$, and $\hat{q}_3(X;\theta)$, corresponding to the probabilities that a pixel belongs to a dog, a cat, or the background, respectively.
From the statistical model, and assuming that the samples (x_i, y_i) are independent and identically distributed, a loss function can be derived to maximize the likelihood of the model given the observed dataset d:

$$\begin{aligned}
\theta_{\text{MLE}} &= \arg\max_\theta p(\mathbf{d} \mid \theta) \\
&= \arg\max_\theta p((x_1, y_1), \ldots, (x_N, y_N) \mid \theta) \\
&= \arg\max_\theta \prod_{i=1}^{N} p(x_i, y_i \mid \theta) \\
&= \arg\max_\theta \prod_{i=1}^{N} p(y_i \mid x_i, \theta) \\
&= \arg\max_\theta \sum_{i=1}^{N} \log(p(y_i \mid x_i, \theta)) \\
&= \arg\min_\theta \sum_{i=1}^{N} -\log(p(y_i \mid x_i, \theta)) \\
&= \arg\min_\theta \mathcal{L}(\theta, \mathbf{d})
\end{aligned}$$

For the first task with Gaussian likelihood, the loss is the mean squared error:

$$\mathcal{L}(\theta, \mathbf{d}) = \frac{1}{500} \sum_{(x_i,y_i)\in\mathbf{d}} (y_i - \hat{\mu}(x_i;\theta))^2$$

For the second task with a Bernoulli likelihood, the loss is the binary cross-entropy:

$$\mathcal{L}(\theta, \mathbf{d}) = -\frac{1}{500} \sum_{(x_i,y_i)\in\mathbf{d}} y_i \log(\hat{q}(x_i;\theta)) + (1 - y_i) \log(1 - \hat{q}(x_i;\theta))$$

For the third task with categorical likelihood, the loss is the cross-entropy:

$$\mathcal{L}(\theta, \mathbf{d}) = -\frac{1}{500} \sum_{(x_i,y_i)\in\mathbf{d}} \frac{1}{32 \times 32} \sum_{u=0}^{31} \sum_{v=0}^{31} \sum_{c=0}^{2} y_{i,c,uv} \log(\hat{q}_c(x_i;\theta))$$

Regarding the output layer, the linear projection

$$W_y h + b_y$$

must map the output h of the preceding layer of the neural network to the desired output dimension.
For the first task:

$$W_y \in \mathbb{R}^{1,d_o} \quad b_y \in \mathbb{R}^1$$

For the second task:

$$W_y \in \mathbb{R}^{1,d_o} \quad b_y \in \mathbb{R}^1$$

For the third task:

$$W_y \in \mathbb{R}^{32,32,3,d_o} \quad b_y \in \mathbb{R}^{32,32,3}$$

The non-linear activation function σ should map the projected output to the valid range of values for y. For the first task, we can choose the identity:

$$\sigma(\cdot) = \cdot$$

For the second task, we can choose a sigmoid activation to map the projected values between 0 and 1:

$$\sigma(\cdot) = \frac{1}{1 + \exp(-\cdot)}$$

For the third task, we can choose the Softmax activation to obtain logits for the three values associated for each pixel j, i:

$$\sigma(y_{j,i,u}) = \frac{\exp(y_{j,i,u})}{\sum_{u=0}^{2} \exp(y_{j,i,u})} \quad \forall i, j, u$$

You want to solve a new task: estimate the age of the dog or the cat in the image. To simplify the problem, you consider the age as an unrestricted real-valued quantity. After some reflection, it becomes clear that directly inferring the age from the image is not a good idea. If the type of animal (cat or dog) were first estimated, it would be much easier to estimate the animal's age from the image and this information.

(d) Formulate the task in the context of supervised learning and explain all the elements appearing in $p_\theta(y \mid x)$. What mathematical quantity is estimated by the neural network?

Solution. $y \sim Y$ is the age of the animal in the picture ($Y \in \mathbb{R}$).

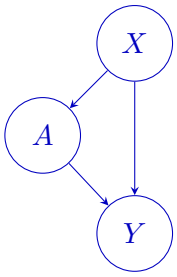
$x \sim X$ is the picture ($X \in \mathbb{R}^{3,32,32}$).

θ is the parameters of the model.

The problem is a regression problem. The neural network should give an estimate of $\mathbb{E}\{Y \mid X\}$.

(e) Draw a Bayesian network modelling the image, the type of animal and its age. Find a new expression for $p(y \mid x)$ that makes the effect of the type of the animal in the image explicit.

Solution. Let A denote the random variable describing the type of animal present in the image (let's say $A = 1$ if it is a dog). The animal type A depends on the image X and influences the age Y. Thus, the age Y depends on both the image X and the animal type A.



$$\begin{aligned} p(y \mid x) &= \sum_a p(y \mid A = a, x) p(A = a \mid x) \\ &= p(y \mid A = 1, x) p(A = 1 \mid x) + p(y \mid A = 0, x)(1 - p(A = 1 \mid x)) \end{aligned}$$

(f) Consider a large neural network designed to solve the task, divided into multiple submodules, each implemented as a separate neural network. Use the image classification model developed above as one of these submodules. Define the inputs and outputs of the additional submodules required to solve the task. Relate the resulting architecture to the mathematical expression derived above. Which submodule estimates each of the distributions appearing in this expression?

Hint: Derive an expression for $\mathbb{E}\{Y \mid X\}$ using the Bayesian network, explicitly accounting for the animal type A.

Solution. The neural network gives an estimate of the conditional expectation:

$$\begin{aligned}\mathbb{E}\{Y \mid X\} &= \mathbb{E}\{\mathbb{E}\{Y \mid X, A\}\} \\ &= \sum_a \mathbb{E}\{Y \mid X, A = a\} p(A = a \mid X) \\ &= \mathbb{E}\{Y \mid X, A = 1\} p(A = 1 \mid X) + \mathbb{E}\{Y \mid X, A = 0\}(1 - p(A = 1 \mid X))\end{aligned}$$

The expression can also be obtained directly by applying the expectation operator $\mathbb{E}_{Y|X}\{\cdot\}$ to both sides of the expression derived in (e).

The expected age is therefore a weighted mean of the age inferred under the assumption that the image contains a dog and the age inferred under the assumption that it contains a cat. Each age is weighted by the corresponding probability.

The first submodule is the image classification model developed previously. With parameters ϕ, it takes an image x as input and estimates the probability that the image contains a dog:

$$\hat{q}(x; \phi) \approx p(A = 1 \mid x) = 1 - p(A = 0 \mid x)$$

The second submodule, with parameters ξ, takes the image x as input and estimates the expected age of the animal under the assumption that it is a dog:

$$\hat{\mu}_d(x; \xi) \approx \mathbb{E}\{Y \mid x, A = 1\}$$

A third submodule, with parameters ψ, takes the image X as input and estimates the expected age under the assumption that it is a cat:

$$\hat{\mu}_c(x; \psi) \approx \mathbb{E}\{Y \mid x, A = 0\}$$

The final estimated age can then be reconstructed from the outputs of these submodules.

Exercise 11 Shallow network expressiveness (Exam, August 2026, adapted)
We consider a neural network for binary classification that takes as input a 2D point $(x_1, x_2) \in \mathbb{R}^2$ and outputs a probability $\hat{y} = f(x_1, x_2) \in [0, 1]$. The network has one hidden layer with two units and is defined as

$$\begin{aligned}h_0 &= \mathrm{ReLU}(x_1 - 1), \\ h_1 &= \mathrm{ReLU}(x_2 - 2), \\ \hat{y} &= \sigma(h_0 + h_1 - 2),\end{aligned}$$

where

$$\mathrm{ReLU}(z) = \max(0, z)$$

is the rectified linear unit and

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

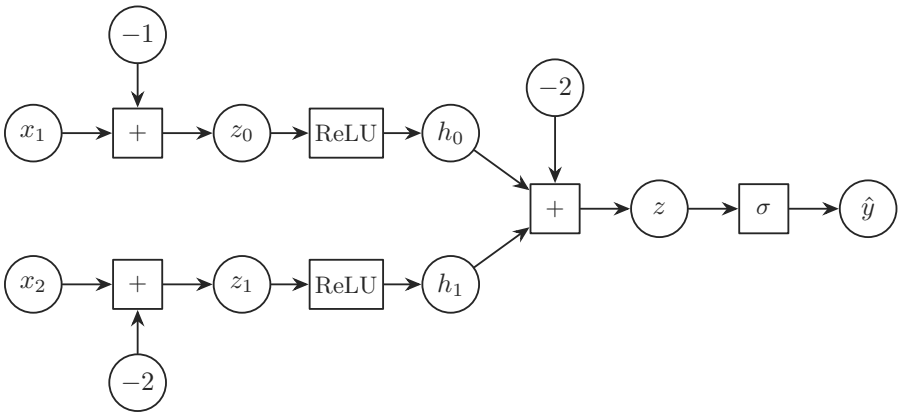
is the sigmoid function.

The predicted class is

$$\hat{c} = \begin{cases} 1, & \text{if } \hat{y} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Draw the computation graph representing the neural network. Use circled nodes for variables (inputs, outputs, and intermediate values) and squared nodes for operations (addition, ReLU, and sigmoid).

Solution.



(b) The ReLU activation functions divide the input space into several regions, depending on whether each hidden unit is active ($h_j > 0$) or inactive ($h_j = 0$).

i. Identify and characterize these regions in terms of conditions on x_1 and x_2.

Solution. The two ReLU units are characterized by

$$\begin{aligned} x_1 > 1 &\implies h_0 = x_1 - 1 > 0, \\ x_1 \leq 1 &\implies h_0 = 0, \end{aligned}$$

and

$$\begin{aligned} x_2 > 2 &\implies h_1 = x_2 - 2 > 0, \\ x_2 \leq 2 &\implies h_1 = 0. \end{aligned}$$

Therefore, four regions can be distinguished:

1. $$\begin{cases} x_1 > 1 \\ x_2 > 2 \end{cases} \implies \begin{cases} h_0 = x_1 - 1 \\ h_1 = x_2 - 2 \end{cases}$$
2. $$\begin{cases} x_1 > 1 \\ x_2 \leq 2 \end{cases} \implies \begin{cases} h_0 = x_1 - 1 \\ h_1 = 0 \end{cases}$$
3. $$\begin{cases} x_1 \leq 1 \\ x_2 > 2 \end{cases} \implies \begin{cases} h_0 = 0 \\ h_1 = x_2 - 2 \end{cases}$$
4. $$\begin{cases} x_1 \leq 1 \\ x_2 \leq 2 \end{cases} \implies \begin{cases} h_0 = 0 \\ h_1 = 0 \end{cases}$$

ii. For each region, give a simplified expression for the output probability $\hat{y}$ as a function of x_1 and x_2.

Solution.

1. For $x_1 > 1$ and $x_2 > 2$,

$$\begin{aligned} \hat{y} &= \sigma(h_0 + h_1 - 2) \\ &= \sigma((x_1 - 1) + (x_2 - 2) - 2) \\ &= \sigma(x_1 + x_2 - 5) \\ &= \boxed{\frac{1}{1 + e^{-(x_1 + x_2 - 5)}}}. \end{aligned}$$

2. For $x_1 > 1$ and $x_2 \leq 2$,

$$\begin{aligned}\hat{y} &= \sigma(h_0 + h_1 - 2) \\ &= \sigma((x_1 - 1) + 0 - 2) \\ &= \sigma(x_1 - 3) \\ &= \boxed{\frac{1}{1 + e^{-(x_1 - 3)}}}.\end{aligned}$$

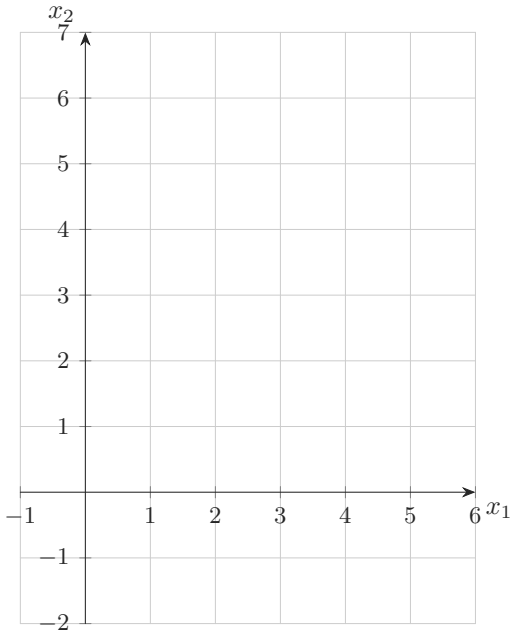
3. For $x_1 \leq 1$ and $x_2 > 2$,

$$\begin{aligned}\hat{y} &= \sigma(h_0 + h_1 - 2) \\ &= \sigma(0 + (x_2 - 2) - 2) \\ &= \sigma(x_2 - 4) \\ &= \boxed{\frac{1}{1 + e^{-(x_2 - 4)}}}.\end{aligned}$$

4. For $x_1 \leq 1$ and $x_2 \leq 2$,

$$\begin{aligned}\hat{y} &= \sigma(0 + 0 - 2) \\ &= \sigma(-2) \\ &= \boxed{\frac{1}{1 + e^{2}}}.\end{aligned}$$

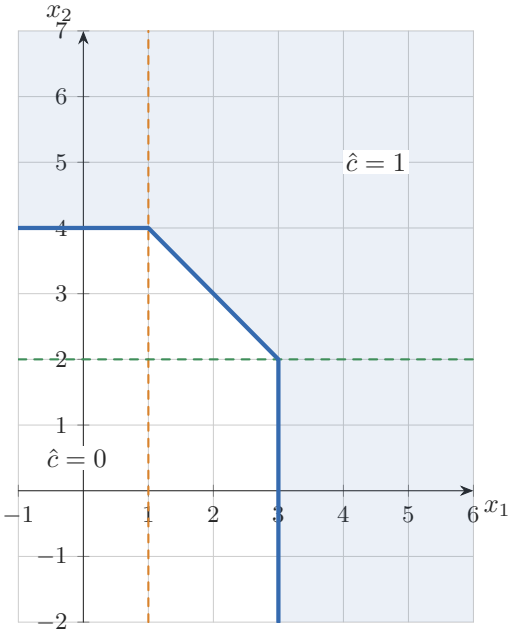
(c) Plot the decision boundary $\hat{y} = 0.5$ of the classifier on the grid below. Clearly indicate which region(s) are classified as class 1 and which are classified as class 0.



Solution. Since the sigmoid is increasing and $\sigma(0) = 0.5$, we have $\hat{y} \geq 0.5$ if and only if $h_0 + h_1 - 2 \geq 0$. In each region of (b):

1. for $x_1 > 1$ and $x_2 > 2$, $\hat{c} = 1$ if and only if $x_1 + x_2 \geq 5$;
2. for $x_1 > 1$ and $x_2 \leq 2$, $\hat{c} = 1$ if and only if $x_1 \geq 3$;
3. for $x_1 \leq 1$ and $x_2 > 2$, $\hat{c} = 1$ if and only if $x_2 \geq 4$;
4. for $x_1 \leq 1$ and $x_2 \leq 2$, $\hat{y} = \sigma(-2) < 0.5$, so $\hat{c} = 0$ everywhere.

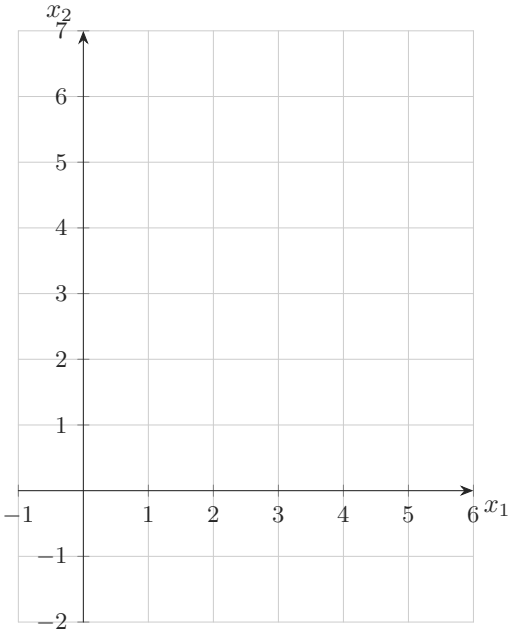
The decision boundary is the broken line from $(-\infty, 4)$ to $(1, 4)$, $(3, 2)$ and $(3, -\infty)$; class 1 (shaded) lies above and to the right of it, class 0 below and to the left. The joints of the hidden units are dashed.



(d) We now consider a modified network with a third hidden unit:

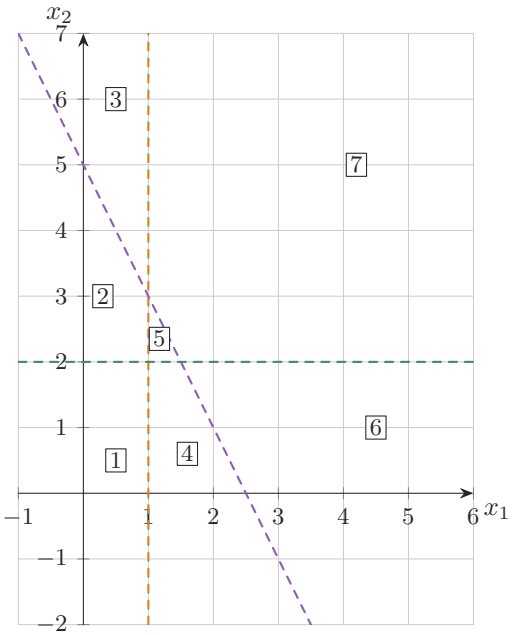
$$
\begin{aligned}
h_0 &= \mathrm{ReLU}(x_1 - 1), \\
h_1 &= \mathrm{ReLU}(x_2 - 2), \\
h_2 &= \mathrm{ReLU}(-2x_1 - x_2 + 5), \\
\hat{y} &= \sigma(h_0 + h_1 + h_2 - 2).
\end{aligned}
$$

On the grid below, plot the new decision boundary and compare it to the original one. What is the effect of adding h_2?



Solution. The new unit is active when $2x_1 + x_2 < 5$. The three joints $x_1 = 1$, $x_2 = 2$ and

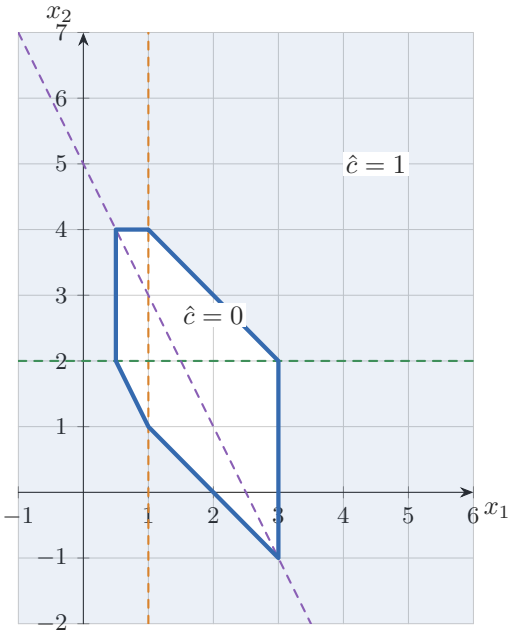
$2x_1 + x_2 = 5$ cut the plane into seven linear regions; the eighth activation pattern, with all units inactive, cannot occur, since $x_1 \leq 1$ and $x_2 \leq 2$ give $2x_1 + x_2 \leq 4 < 5$.



With $z = h_0 + h_1 + h_2 - 2$, the class is 1 where $z \geq 0$:

1. $x_1 \leq 1$, $x_2 \leq 2$ (h_2 only): $z = -2x_1 - x_2 + 3$, class 1 where $2x_1 + x_2 \leq 3$;
2. $x_1 \leq 1$, $x_2 > 2$, $2x_1 + x_2 < 5$ (h_1, h_2): $z = -2x_1 + 1$, class 1 where $x_1 \leq \frac{1}{2}$;
3. $x_1 \leq 1$, $x_2 > 2$, $2x_1 + x_2 \geq 5$ (h_1 only): $z = x_2 - 4$, class 1 where $x_2 \geq 4$;
4. $x_1 > 1$, $x_2 \leq 2$, $2x_1 + x_2 < 5$ (h_0, h_2): $z = -x_1 - x_2 + 2$, class 1 where $x_1 + x_2 \leq 2$;
5. $x_1 > 1$, $x_2 > 2$, $2x_1 + x_2 < 5$ (all three): $z = -x_1 < 0$, class 0 everywhere;
6. $x_1 > 1$, $x_2 \leq 2$, $2x_1 + x_2 \geq 5$ (h_0 only): $z = x_1 - 3$, class 1 where $x_1 \geq 3$;
7. $x_1 > 1$, $x_2 > 2$, $2x_1 + x_2 \geq 5$ (h_0, h_1): $z = x_1 + x_2 - 5$, class 1 where $x_1 + x_2 \geq 5$.

Class 0 is now the bounded hexagon with vertices $(1, 1)$, $(\frac{1}{2}, 2)$, $(\frac{1}{2}, 4)$, $(1, 4)$, $(3, 2)$ and $(3, -1)$, and everything else is class 1 (shaded), including the origin, where $z = 3$.



Where h_2 is inactive ($2x_1 + x_2 \geq 5$, regions 3, 6 and 7), the boundary is unchanged. Where it is active, it adds $-2x_1 - x_2 + 5$ to z and changes the decision: the region of class 0, which was unbounded and contained the origin, becomes a bounded hexagon. One more unit adds one more joint, and with it more linear regions and a more complex decision boundary.

Exercise 12 Neural network parameters

Sandrine and Karl, two long-time data scientist friends, each decide to design a neural network to solve a challenging task: compressing a high-definition color image in a HD 720p resolution (1280×720 pixels).

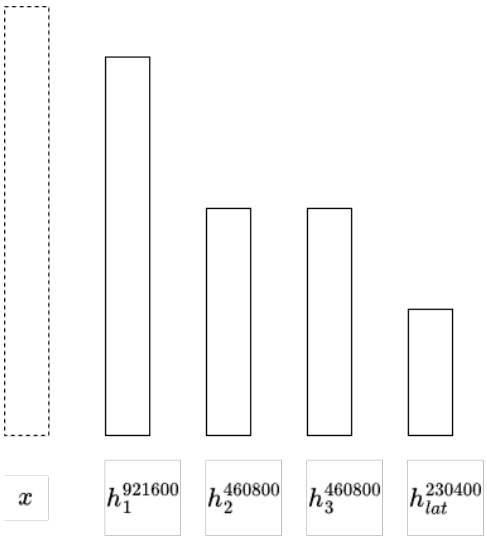
To do so, both decide to use an encoder network and a decoder network. The first network, the *encoder*, transforms the input image into a low-dimensional latent representation which constitutes the compressed representation. The second network, the *decoder*, reconstructs the original image as faithfully as possible from this low-dimensional representation.

Sandrine decides to use a multilayer perceptron with ReLU nonlinear activations for her encoder. Each block i of her encoder therefore consists of a linear projection followed by the nonlinear activation:

$$h_i = \sigma\left(W_i h_{i-1} + b_i\right).$$

The HD color image is represented using three channels corresponding to the red, green, and blue components. Sandrine flattens the image into a single vector before feeding it to the network.

The dimensions of the different units throughout Sandrine's encoder are given below:



(a) How many trainable parameters does Sandrine's encoder contain? Is this network realistic to implement on a standard laptop? Assume that each parameter is encoded using 32 bits.

Solution. The dimension of W_i and b_i for each layer are:

1. $W_1^{921600 \times 2764800}$ b_1^{921600}
2. $W_2^{460800 \times 921600}$ b_2^{460800}
3. $W_3^{460800 \times 460800}$ b_3^{460800}
4. $W_4^{230400 \times 460800}$ b_4^{230400}

The total number of parameters is:

$$\begin{aligned} N_{\text{param}} &= 921600 \times 2764800 + 921600 + 460800 \times 921600 + 460800 \\ &\quad + 460800 \times 460800 + 460800 + 230400 \times 460800 + 230400 \\ &= \boxed{3\,291\,219\,993\,600}. \end{aligned}$$

Thus, using 32 bits (4 bytes) per parameter, the memory required to store the parameters is

$$\begin{aligned} M &= 3\,291\,219\,993\,600 \times 4 \text{ bytes} = \frac{13\,164\,879\,974\,400}{2^{40}} \text{ TiB} \\ &\approx \boxed{11.97 \text{ TiB}}. \end{aligned}$$

Clearly, Sandrine's encoder is infeasible.

(b) What is the compression ratio achieved by Sandrine's encoder?

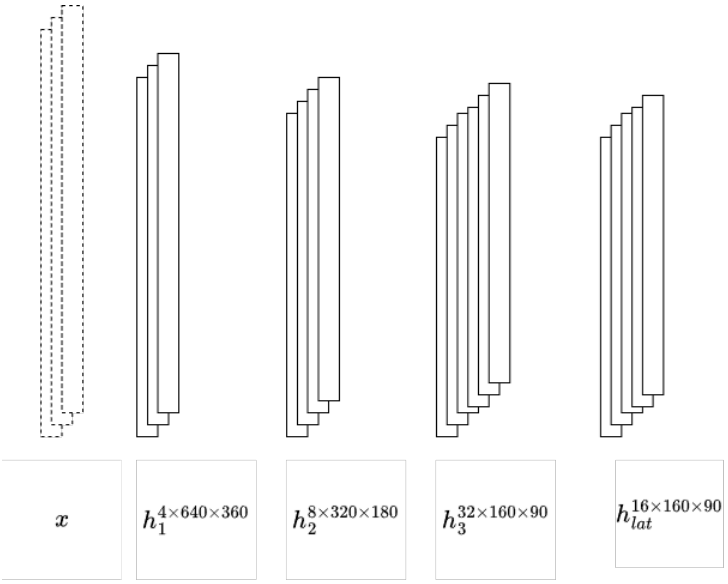
Solution.

$$\eta = 1 - \frac{230400}{3 \times 1280 \times 720} = 92\%$$

Karl decides to use a convolutional neural network, also with ReLU nonlinear activations for his encoder.

Each block of the network consists of a convolutional layer with 3×3 kernels, followed by a ReLU activation and, in every block but the last, a max-pooling layer with a pooling area of 2×2. Before being passed through each block, the input feature maps of each block are padded with a one-pixel-wide border of zeros.

The HD color image is still represented using three channels corresponding to the red, green, and blue components. The dimensions of the different units throughout Karl's encoder are given below:



(c) How many parameters does Karl's encoder contain? Is this network realistic to implement on a standard laptop? Assume that each parameter is encoded using 32 bits.

Solution. Let the input of a block be

$$h \in \mathbb{R}^{a \times b \times c},$$

where a denotes the number of input channels, and b and c denote the height and width of the feature maps, respectively.

Before the convolution, the input is padded with a border of zeros of width 1. The padded input therefore has shape

$$h_p \in \mathbb{R}^{a\times(b+2)\times(c+2)}.$$

A convolutional kernel of spatial size 3×3 must span all a input channels. Consequently, each kernel has shape

$$a \times 3 \times 3$$

and therefore contains

$$9a$$

trainable weights. If the convolutional layer contains k kernels, the total number of weights is

$$\boxed{9ak}.$$

As each kernel also has a bias, the total number of trainable parameters is

$$\boxed{9ak + k = k(9a + 1)}.$$

Because the input is padded by one pixel on each side and the convolution uses a 3×3 kernel, the spatial dimensions are preserved:

$$(b + 2) - 3 + 1 = b, \qquad (c + 2) - 3 + 1 = c.$$

Therefore, the output of the convolutional layer has shape

$$k \times b \times c.$$

The ReLU does not modify the dimensions. Finally, the 2×2 max-pooling layer reduces each spatial dimension by a factor of 2. Assuming b and c are even, the output of the block has shape

$$\boxed{k \times \frac{b}{2} \times \frac{c}{2}},$$

except for the last block, which has no pooling and keeps the shape $k \times b \times c$. The number of kernels and their shapes throughout the network are:

1. 4 kernels of shape $3 \times 3 \times 3$,
2. 8 kernels of shape $4 \times 3 \times 3$,
3. 32 kernels of shape $8 \times 3 \times 3$,
4. 16 kernels of shape $32 \times 3 \times 3$.

The total number of parameters is therefore

$$\begin{aligned} N_{\text{param}} &= 4(3 \times 3 \times 3 + 1) + 8(4 \times 3 \times 3 + 1) \\ &\quad + 32(8 \times 3 \times 3 + 1) + 16(32 \times 3 \times 3 + 1) \\ &= 4(28) + 8(37) + 32(73) + 16(289) \\ &= 112 + 296 + 2336 + 4624 \\ &= \boxed{7368}. \end{aligned}$$

Thus, using 32 bits per parameter, the memory required to store the parameters is

$$\begin{aligned} M &= 7368 \times 4 \text{ bytes} = 29\,472 \text{ bytes} \\ &= \frac{29\,472}{2^{10}} \text{ KiB} \\ &\approx \boxed{28.8 \text{ KiB}}. \end{aligned}$$

Karl's encoder clearly fits in a laptop.

(d) What is the compression ratio achieved by Karl's encoder?

Solution.

$$\eta = 1 - \frac{16 \times 160 \times 90}{3 \times 1280 \times 720} = 92\%$$

Exercise 13 Learning to play Pacman (Exam, August 2020)
You observe a Grandmaster agent playing Pacman. How can you use the moves you observe to train your own agent?

(a) Describe formally the data you would collect, the inference problem you would consider, and how you would solve it.

Solution. At each step t of the games we watch, the Grandmaster perceives e_t and plays $a_t \in \mathcal{A} = \{\text{North}, \text{South}, \text{East}, \text{West}, \text{Stop}\}$; Pacman's world is fully observable, so $e_t = s_t$. Each step gives an input-output pair $(\mathbf{x}_i, y_i)$, with features $\mathbf{x}_i = g(e_t) \in \mathbb{R}^d$ computed from the game state (positions of Pacman, the ghosts and the food, distances, walls around Pacman, ...) and the label $y_i = a_t$. Collecting N steps gives a dataset $\mathbf{d} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$. The inference problem is a classification problem with $|\mathcal{A}|$ classes: we look for a model $P_\theta(y \mid \mathbf{x})$ of the Grandmaster's choice given the features, whose parameters θ maximize the likelihood of $\mathbf{d}$,

$$\theta^* = \arg\max_\theta \prod_{i=1}^{N} P_\theta(y_i \mid \mathbf{x}_i).$$

Since there is no closed-form solution for a neural network, we minimize the negative log-likelihood by (stochastic) gradient descent. Our agent is then the simple reflex agent $\pi_\theta(e_t) = \arg\max_{a \in \mathcal{A}} P_{\theta^*}(a \mid g(e_t))$, whose condition-action rules are learned from the Grandmaster rather than written by hand.

(b) How would you design a neural network to control your agent? Define mathematically the neural network architecture, its inputs, its outputs, its parameters, as well as the loss you would use to train it.

Solution. A multilayer perceptron with input $\mathbf{x} = g(e_t) \in \mathbb{R}^d$, K hidden layers and $C = |\mathcal{A}| = 5$ outputs:

$$\begin{aligned} \mathbf{h}_0 &= \mathbf{x}, \\ \mathbf{h}_k &= \text{ReLU}(\mathbf{W}_k \mathbf{h}_{k-1} + \mathbf{b}_k), \quad k = 1, \dots, K, \\ f(\mathbf{x}; \theta) &= \text{Softmax}(\mathbf{W}_y \mathbf{h}_K + \mathbf{b}_y), \end{aligned}$$

with $\mathbf{W}_k \in \mathbb{R}^{q_k \times q_{k-1}}$, $\mathbf{b}_k \in \mathbb{R}^{q_k}$ ($q_0 = d$), $\mathbf{W}_y \in \mathbb{R}^{C \times q_K}$ and $\mathbf{b}_y \in \mathbb{R}^C$, so that $\theta = \{\mathbf{W}_1, \mathbf{b}_1, \dots, \mathbf{W}_K, \mathbf{b}_K, \mathbf{W}_y, \mathbf{b}_y\}$. The output $f_a(\mathbf{x}; \theta)$ is the probability $P_\theta(a \mid \mathbf{x})$ that the Grandmaster plays a. Numbering the actions $0, \dots, C-1$, the loss is the cross-entropy, the negative log-likelihood of $\mathbf{d}$:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \log f_{y_i}(\mathbf{x}_i; \theta).$$

Illegal actions (into a wall) can be masked by setting their probability to zero before taking the arg max.

(c) Discuss the expected performance of the resulting agent when 1. the Grandmaster agent is optimal, and 2. the Grandmaster agent is suboptimal.

Solution. 1. If the Grandmaster is optimal, our agent is at best as good as the Grandmaster, and in practice worse. It only imitates the Grandmaster on the states the Grandmaster visits: once a small error leads it to a state that never appears in $\mathbf{d}$, its predictions there are unreliable and errors can compound over the rest of the game. More data, from varied situations, and features g that generalize across mazes reduce this effect. 2. If the Grandmaster is suboptimal, our agent learns its mistakes along with its good moves, since the loss only rewards agreeing with the Grandmaster. Nothing in the data tells which moves were good: doing better requires a signal about the outcome of the moves, such as the score, which is the setting of reinforcement learning.

Quiz

1. We build a MLP with 2 hidden layers of 32 units, ReLU activations and squared error loss. We discard the biases from our linear layers. We know that our input is a vector of 2 elements and the output is a scalar. How many parameters do we have in the network?
   - ◯ 1024.
   - ◉ 1120.
   - ◯ 1185.
   - ◯ Another value than the ones proposed above.
2. We add biases in the network proposed above. What is the number of parameters of the network?
   - ◯ Another value than the ones proposed below.
   - ◯ 1089.
   - ◉ 1185.
   - ◯ 2048.
   - ◯ 2240.
3. We have an image of shape (H, W), we want to pass it through a layer which reduces each dimension by a constant K, *i.e.* produces an output image of shape $(H-K, W-K)$.
   - ◯ It can be done with a convolution layer with a kernel (K, K) or a linear layer with weight matrix $(HW, (H-K)(W-K))$.
   - ◉ It can be done with a convolution layer with a kernel $(K+1, K+1)$ or a linear layer with weight matrix $(HW, (H-K)(W-K))$.
   - ◯ It can be done with a convolution layer with a kernel $(K-1, K-1)$ or a linear layer with weight matrix (HW, KK).
   - ◯ It can be done with a convolution layer with a kernel $(K+1, K+1)$ or a linear layer with weight matrix (HW, KK).
4. From the previous question, we double the number of input pixels by setting the shape of the input image to (H', W) with $H' = 2H$. The latter implies that the output image should be of shape $(H'-K, W-K)$. If we only look at weight parameters (no biases)...
   - ◯ The number of parameters of both the linear and convolution layer are doubled.
   - ◯ The linear layer will have twice more parameters and the convolution twice less.
   - ◉ The linear layer will have twice more parameters and the convolution will not change.
   - ◯ Nothing changes neither for the linear layer nor the convolution one.

5. In deep learning, a layer in a multi-layer perceptron is defined as ...

   ◯ $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$, where σ is the standard deviation function.

   ◉ $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$, where σ is an activation function, such as the sigmoid function.

   ◯ $\mathbf{h} = \sigma(\mathbf{W} + \mathbf{x} - \mathbf{b})$, where $\mathbf{W} \in \mathbb{R}^{q \times d}$ is matrix of weights.

   ◯ $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$, where $\mathbf{b} \in \mathbb{R}^d$ is the most likely vector of hidden states given $\mathbf{x}$.

6. Arnaud is trying to perform gradient descent on a function $f(x)$ using the update

$$x_{t+1} := x_t - \frac{\partial f}{\partial x}(x_t).$$

   Is gradient descent guaranteed to converge to the global minimum of f?

   ◯ Yes, since he's updating using the gradient of x.

   ◯ Yes, but not for the reason above.

   ◯ No, since he is updating x in the wrong direction.

   ◉ No, but not for the reason above.